Queen Mary
University of London

EBU5376
Digital Signal Processing

Week 4 – Tutorial 1

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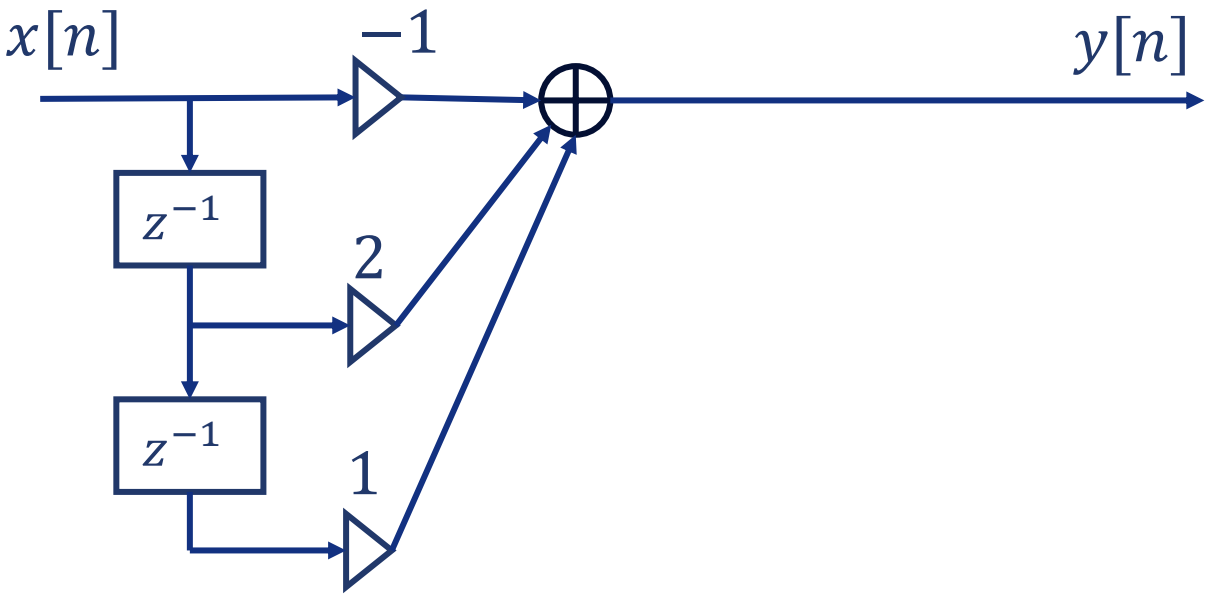
Tutorial Outline

- Block Diagram
- Filter analysis
  - Step Response
  - Amplitude Response
- Evaluation of Amplitude Response
  - Low-pass/High-pass/Band-pass/Band-stop

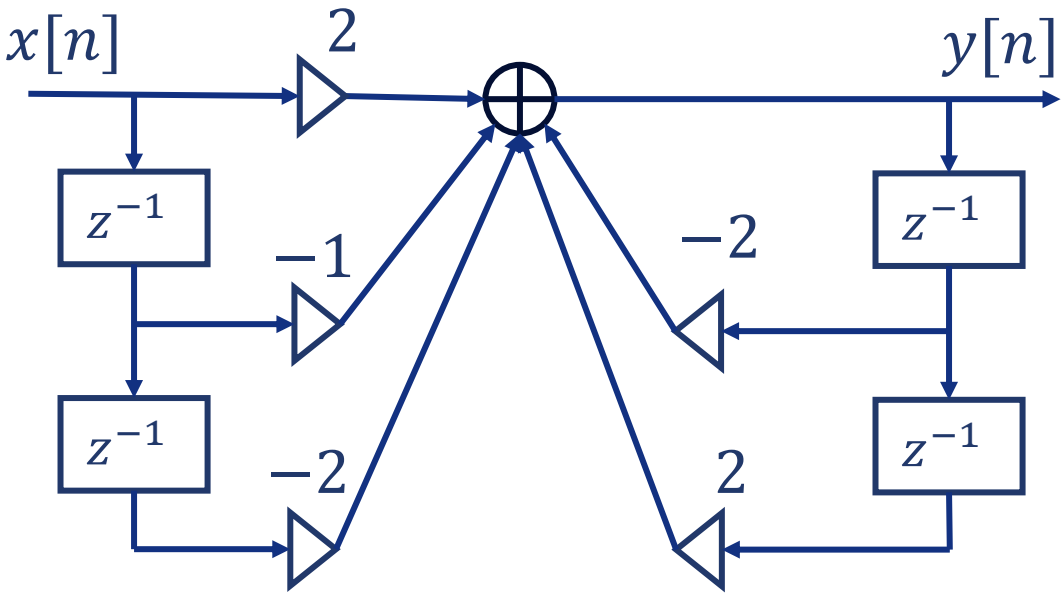
Tutorial Question 1a

Based on the following block diagram, write down the difference equation of the filter.



Tutorial Question 1b

Based on the following block diagram, write down the difference equation of the filter.



Tutorial Question 1c

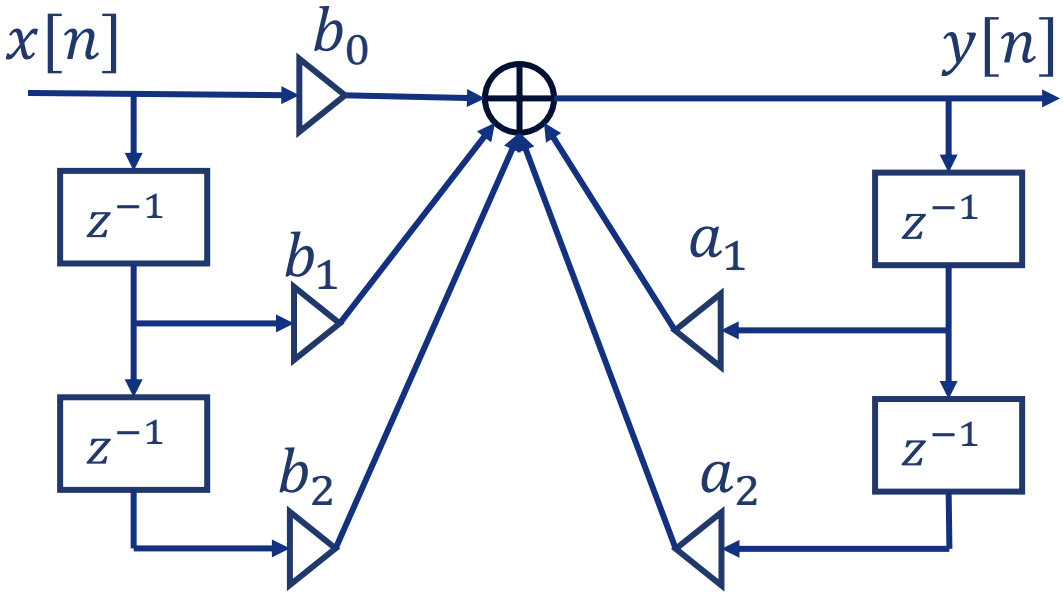
Figure below shows the general block diagram of a filter, where b_0, b_1, b_2, a_1, a_2 are unknown variables. Determine the values of these variables for the following filters:

i. $y[n] = 2x[n] + x[n-1]$

ii. $y[n] = y[n-1]$

iii. $y[n] = y[n-1] + x[n-1]$

iv. Any finite impulse response (FIR) filter.



Tutorial Question 1d

Plot the block diagram of the following filters:

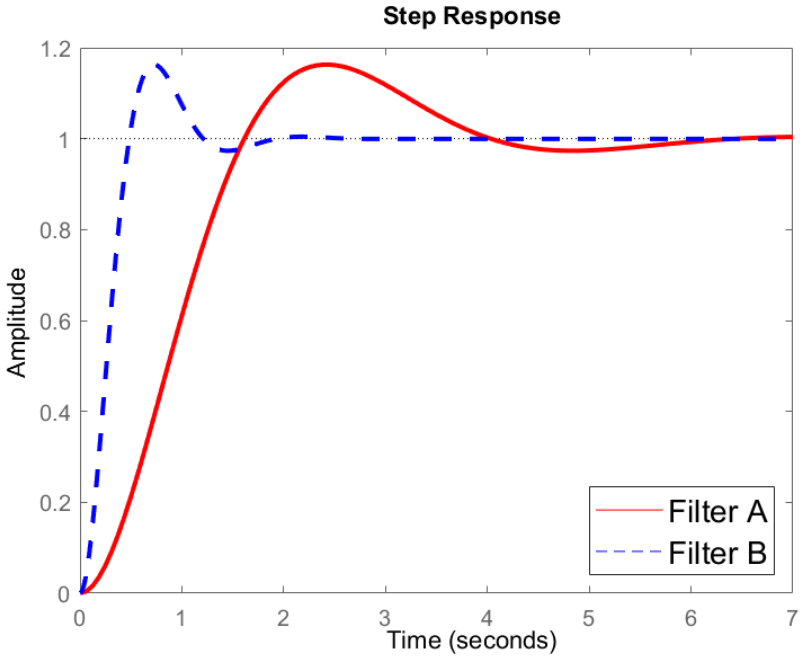
i. $y[n] = x[n] - x[n-1]$

ii. $y[n] = y[n-1]$

iii. $y[n] + 2y[n-1] = x[n-2]$

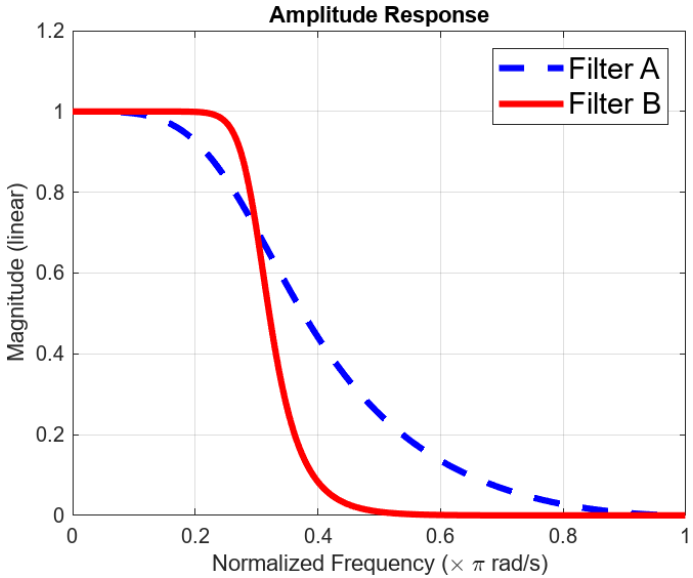
Tutorial Question 2a

Figure below shows the step responses of Filter A and Filter B. Based on the figure, describe and compare the time-domain behaviours of the two filters.



Tutorial Question 2b

Figure below shows the amplitude responses of Filter A and Filter B. Based on the figure, describe and compare the frequency-domain behaviours of the two filters.



Tutorial Question 3a

Consider this filter $y[n] - 0.3y[n-1] = x[n]$

1. Compute the amplitude response

2. Evaluate the amplitude response for frequencies
$$\omega = [0, 0.2\pi, 0.4\pi, 0.6\pi, 0.8\pi, \pi]$$

3. State whether this is a low-pass / band-pass / high-pass / band-stop filter

Tutorial Question 3b

Consider this filter

$$y[n] + 0.5y[n-1] + 0.2y[n-2] = 0.2x[n] - 0.2x[n-1]$$

Determine whether this is a low-pass / band-pass / high-pass / band-stop filter

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